1. MainActivity.kt

```
package com.example.exp1
import android.graphics.Color
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import kotlinx.android.synthetic.main.activity_main.*
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?)
{
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
textView.text = "Hello MAD Lab"
textView.setTextColor(Color.BLUE)
textView.textSize=24f
textView.setBackgroundColor(Color.YELLOW)
}  }
```

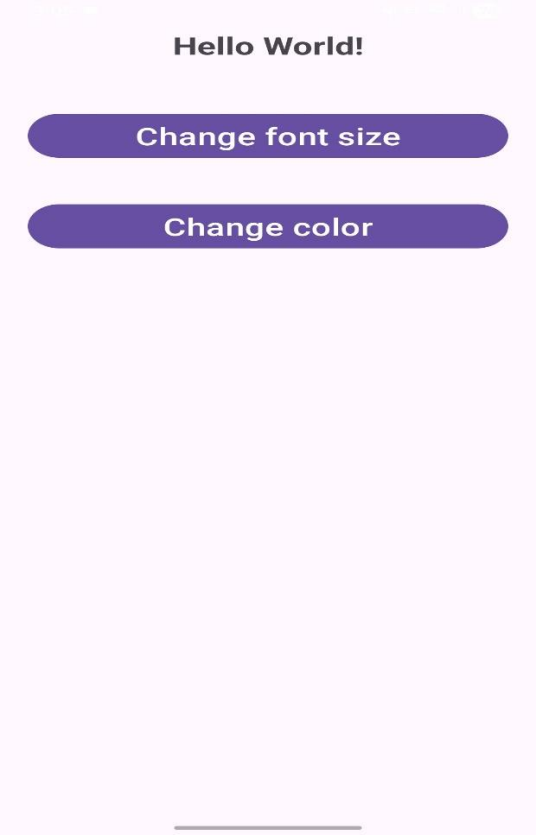
2.activity_main.xml

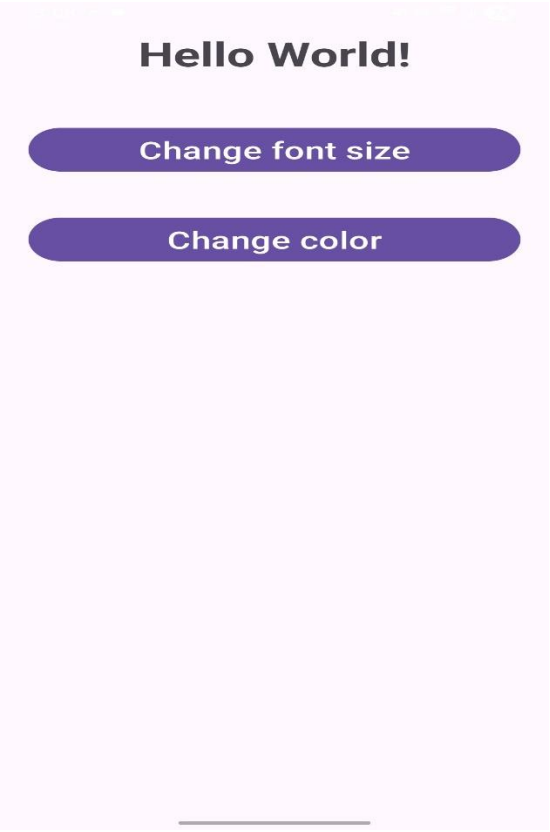
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent">
<TextView
android:id="@+id/textView"
android:layout_centerInParent="true"
android:text="Text"
 android:textSize="18sp"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"/>
</RelativeLayout>
```

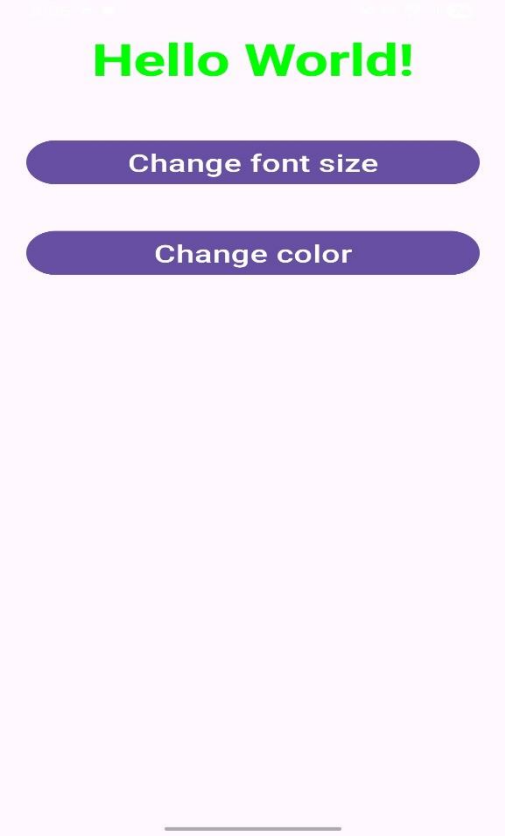
3. AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.exp1">
<application
android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
<activity android:name=".MainActivity">
<intent-filter>
<action android:name="android.intent.action.MAIN"/>
<category android:name="android.intent.category.LAUNCHER"/>
</intent-filter>
</activity>
</application>
</manifest>
```

OUTPUT:


Hello World!
Change font size
Change color


Hello World!
Change font size
Change color


Hello World!
Change font size
Change color

1. MainActivity.kt

```
package com.example.exp2
import android.os.Bundle
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import kotlinx.android.synthetic.main.activity_main.*
class MainActivity : AppCompatActivity() {
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 setContentView(R.layout.activity_main)
 button.setOnClickListener {
Toast.makeText(this, "Button Clicked", Toast.LENGTH_SHORT).show()
}
}
}
```

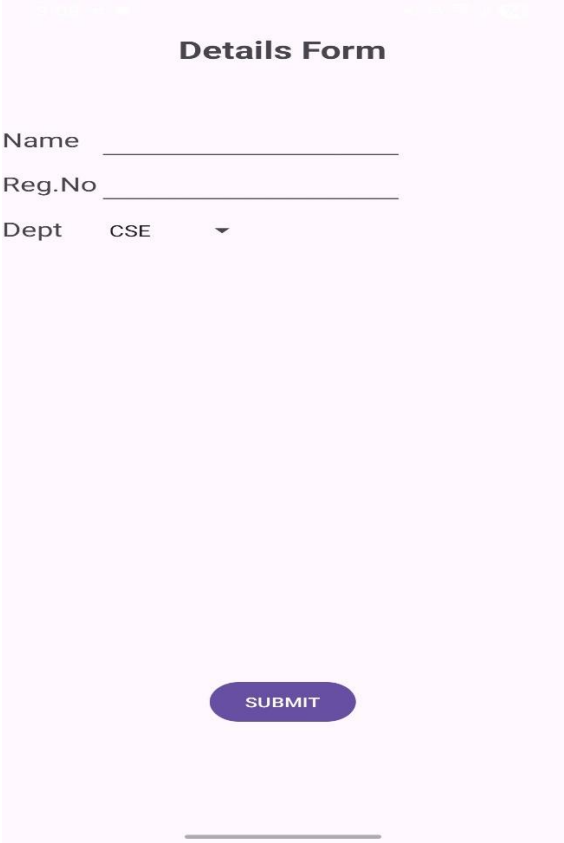
2. activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:gravity="center"
android:layout_width="match_parent"
android:layout_height="match_parent">
<Button
 android:id="@+id/button"
 android:text="Click Me"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"/>
</LinearLayout>
```

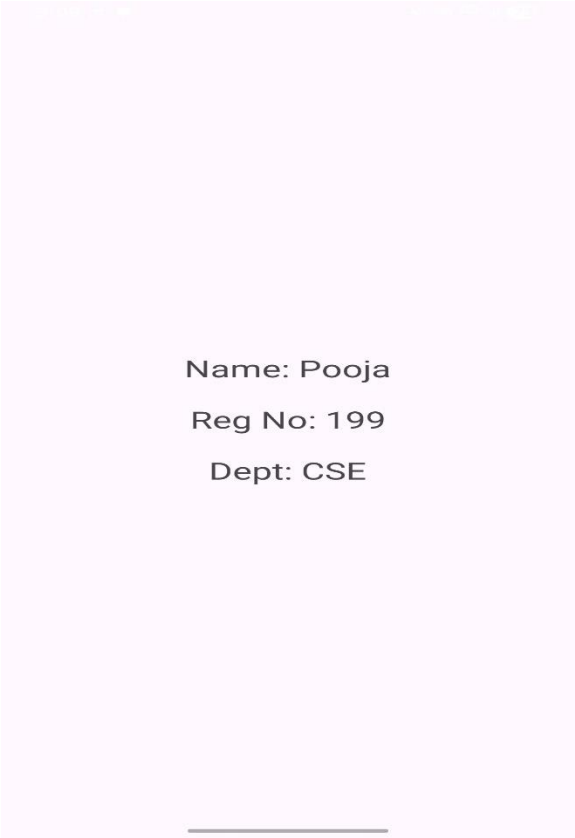
3. AndroidManifest.xml

```
<manifest package="com.example.exp2"
xmlns:android="http://schemas.android.com/apk/res/android">
<application>
<activity android:name=".MainActivity"
<intent-filter>
<action android:name="android.intent.action.MAIN"/>
 <category android:name="android.intent.category.LAUNCHER"/>
</intent-filter>
</activity>
</application>
</manifest>
```

OUTPUT:


Details Form
Name
Reg.No
Dept
CSE
SUBMIT

Details Form
Name
Pooja
Reg.No
199
Dept
CSE
SUBMIT


Name: Pooja
Reg No: 199
Dept: CSE

1. MainActivity.kt

```
package com.example.exp3
import android.content.Intent
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import kotlinx.android.synthetic.main.activity_main.
class MainActivity : AppCompatActivity() {
 override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
button.setOnClickListener {
val intent = Intent(Intent.ACTION_SEND)
intent.type = "message/rfc822"
intent.putExtra(Intent.EXTRA_EMAIL, arrayOf("test@gmail.com"))
 intent.putExtra(Intent.EXTRA_SUBJECT, "Subject")
 intent.putExtra(Intent.EXTRA_TEXT, "Message")
 startActivity(intent)
}  }  }
```

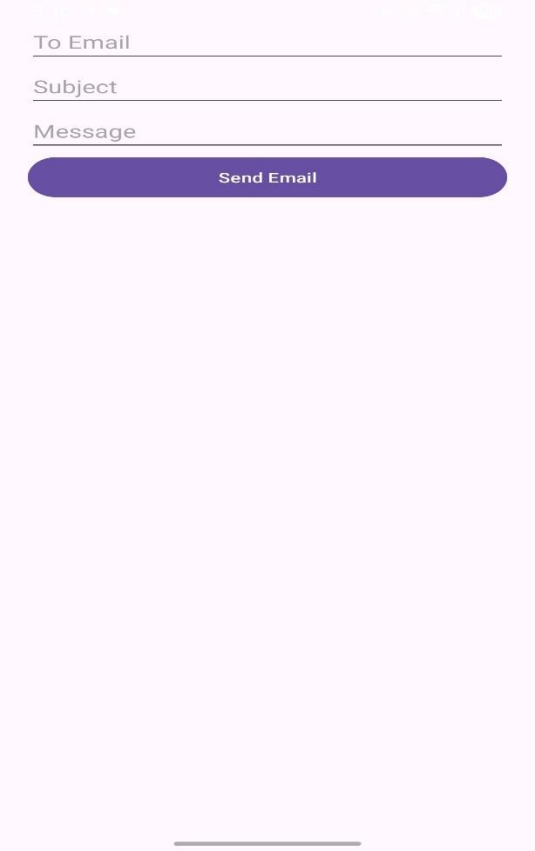
2. activity_main.xml

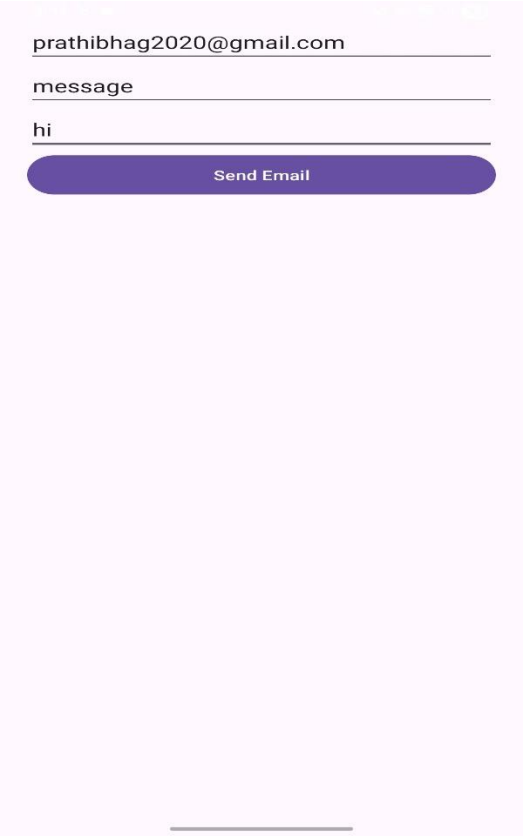
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:gravity="center"
 android:layout_width="match_parent"
 android:layout_height="match_parent">
 <Button
 android:id="@+id/button"
 android:text="Send Email"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"/>
</LinearLayout>
```

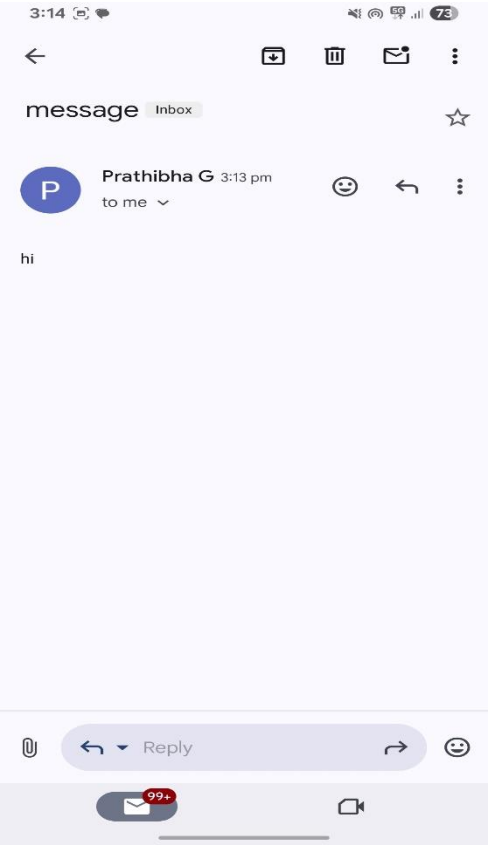
3. AndroidManifest.xml

```
<manifest package="com.example.exp3"
xmlns:android="http://schemas.android.com/apk/res/android">
 <application>
 <activity android:name=".MainActivity"/>
 </application>
</manifest>
```

OUTPUT:


To Email
Subject
Message
Send Email


prathibhag2020@gmail.com
message
hi
Send Email


3:14
message
Inbox
P
Prathibha G 3:13 pm
to me
hi
Reply
99+

1. MainActivity.kt

```
package com.example
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(MyView(this))  }   }
```

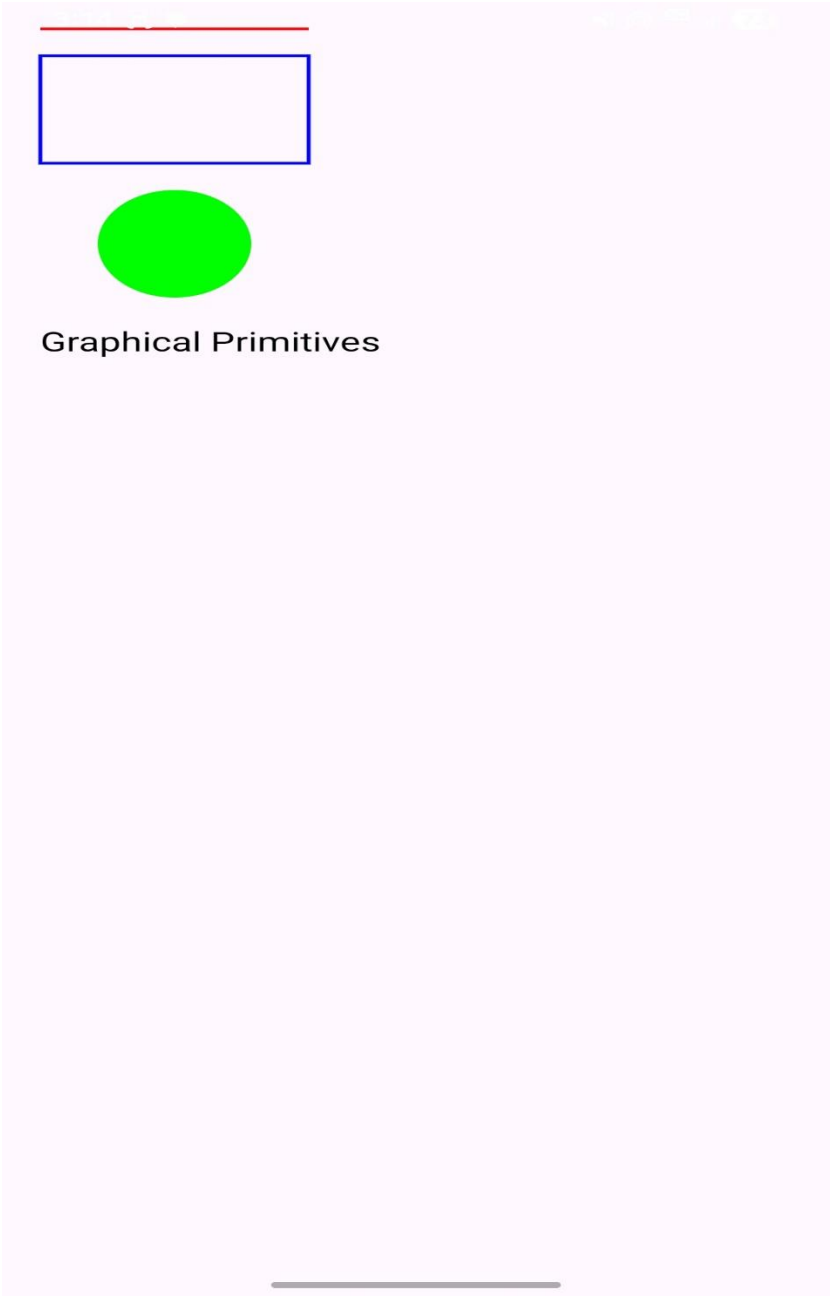
2. MyView.kt

```
package com.example.exp4
import android.content.Context
import android.graphics.*
import android.view.View
class MyView(context: Context) : View(context) {
 override fun onDraw(canvas: Canvas) {
 super.onDraw(canvas)
val paint = Paint()
 paint.color = Color.RED
 canvas.drawCircle(300f, 300f, 100f, paint)
 canvas.drawRect(100f, 500f, 400f, 700f, paint)     }   }
```

3. AndroidManifest.xml

```
<manifest package="com.example.exp4"
 xmlns:android="http://schemas.android.com/apk/res/android">
  <application>
  <activity android:name=".MainActivity">
  <intent-filter>
  <action android:name="android.intent.action.MAIN"/>
  <category android:name="android.intent.category.LAUNCHER"/>
 </intent-filter>    </activity>
 </application>
</manifest>
```

OUTPUT:



1. DBHelper.kt

```
package com.example.exp5
Import android.content.Context
import android.database.sqlite.SQLiteDatabase
import android.database.sqlite.SQLiteOpenHelper
class DBHelper(context: Context) :
SQLiteOpenHelper(context, "StudentDB", null, 1) {
 override fun onCreate(db: SQLiteDatabase) {
 db.execSQL("CREATE TABLE student(id INTEGER PRIMARY KEY,
 name TEXT)")
 }
 override fun onUpgrade(db: SQLiteDatabase, oldV: Int, newV: Int) {
 db.execSQL("DROP TABLE IF EXISTS student")
 onCreate(db)
 }
 }
```

2. MainActivity.kt

```
 package com.example.exp5
 import android.os.Bundle
 import android.widget.Button
 import android.widget.Toast
 import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
 lateinit var dbHelper: DBHelper
 override fun onCreate(savedInstanceState: Bundle?) {
 super.onCreate(savedInstanceState)
 val btn = Button(this)
  btn.text = "Insert Data"
  setContentView(btn)
  dbHelper = DBHelper(this)
```

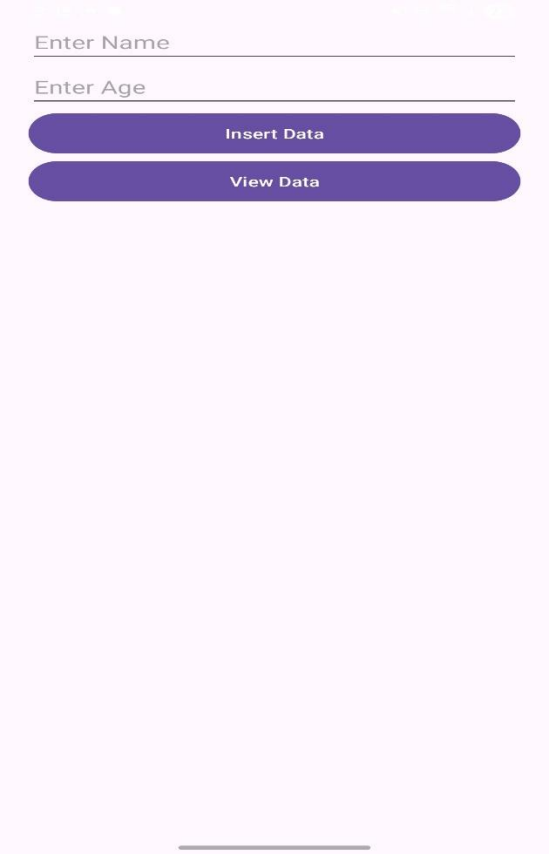
```
btn.setOnClickListener {
val db = dbHelper.writableDatabase
db.execSQL("INSERT INTO student VALUES(1,'Rumi')")
Toast.makeText(this, "Inserted", Toast.LENGTH_SHORT).show()     }     }     }
```

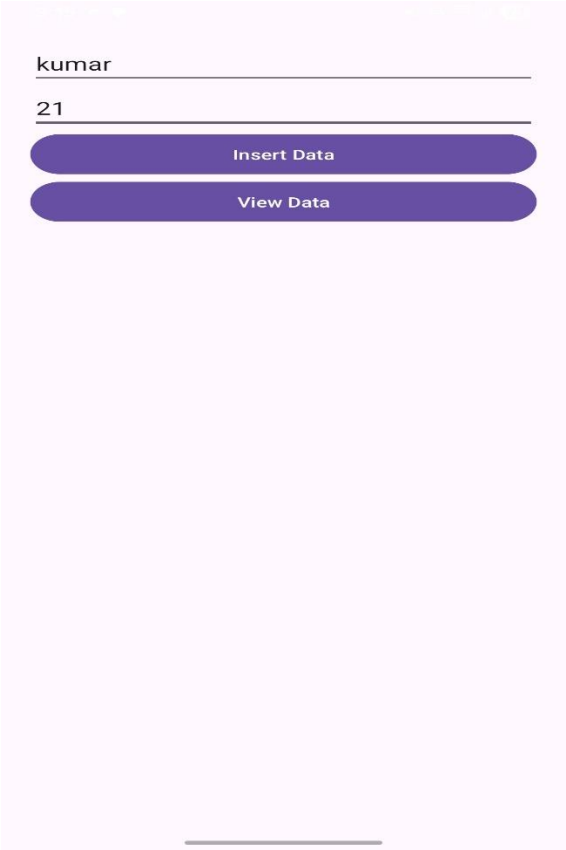
```
package com.example.exp5
import android.os.Bundle
Import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
lateinit var dbHelper: DBHelper
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
val btn = Button(this)
btn.text = "Insert Data"
dbHelper = DBHelper(this)
btn.setOnClickListener {
val db = dbHelper.writableDatabase
db.execSQL("INSERT INTO student VALUES(1,'Rumi')")
Toast.makeText(this, "Inserted", Toast.LENGTH_SHORT).show()        }  }  }
```

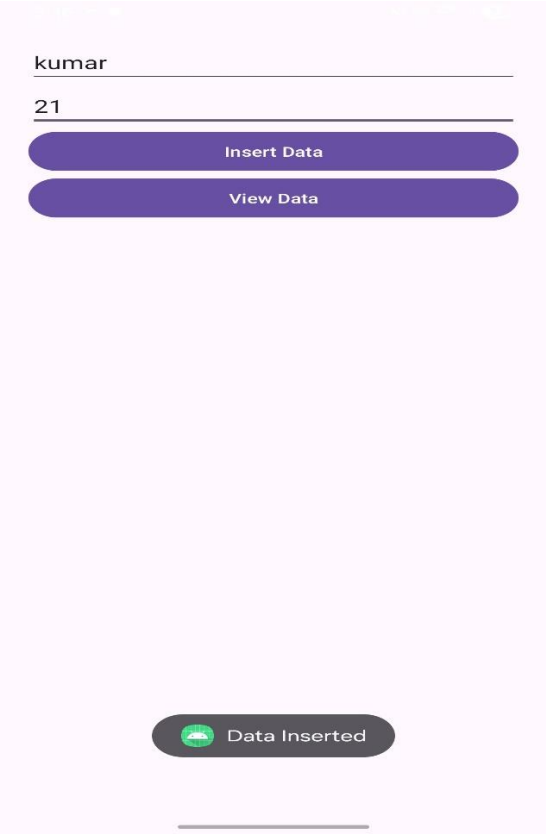
3. AndroidManifest.xml

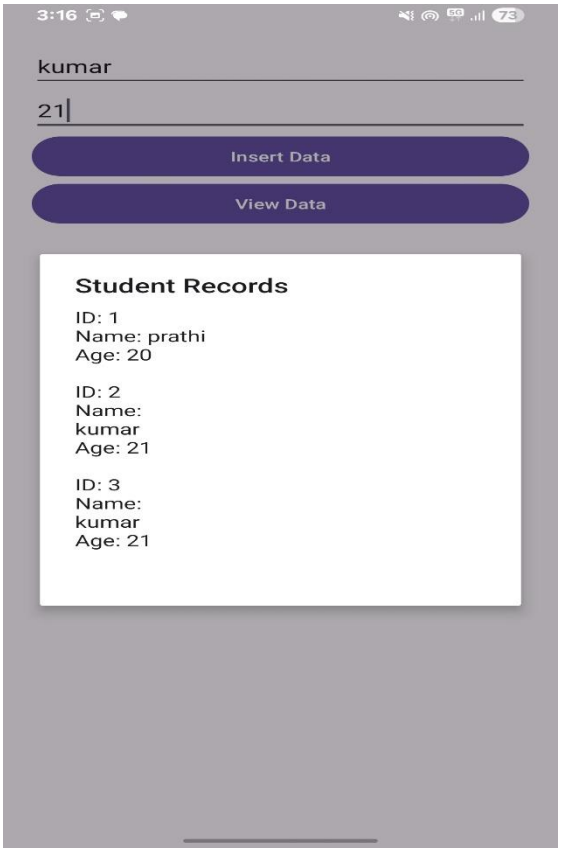
```
<manifest package="com.example.exp5"
xmlns:android="http://schemas.android.com/apk/res/android">
<application>
<activity android:name=".MainActivity"/>
</application>
</manifest>
```

OUTPUT:


Enter Name
Enter Age
Insert Data
View Data


kumar
21
Insert Data
View Data


kumar
21
Insert Data
View Data
Data Inserted


3:16
kumar
21
Insert Data
View Data
Student Records
ID: 1
Name: prathi
Age: 20
ID: 2
Name:
kumar
Age: 21
ID: 3
Name:
kumar
Age: 21

1. MainActivity.kt

```
package com.example.exp6
import android.app.Activity
import android.content.Intent
import android.os.Bundle
import android.speech.RecognizerIntent
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
lateinit var tv: TextView
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
val layout = LinearLayout(this)
layout.orientation = LinearLayout.VERTICAL
val btn = Button(this)
btn.text = "Speak"
tv = TextView(this)
layout.addView(btn)
layout.addView(tv)
setContentView(layout)
btn.setOnClickListener {
val intent = Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH)
startActivityForResult(intent, 1)
}
}
override fun onActivityResult(req: Int, res: Int, data: Intent?) {
super.onActivityResult(req, res, data)
if (req == 1 && res == Activity.RESULT_OK) {
val result = data?.getStringArrayListExtra(RecognizerIntent.EXTRA_RESULTS)
```

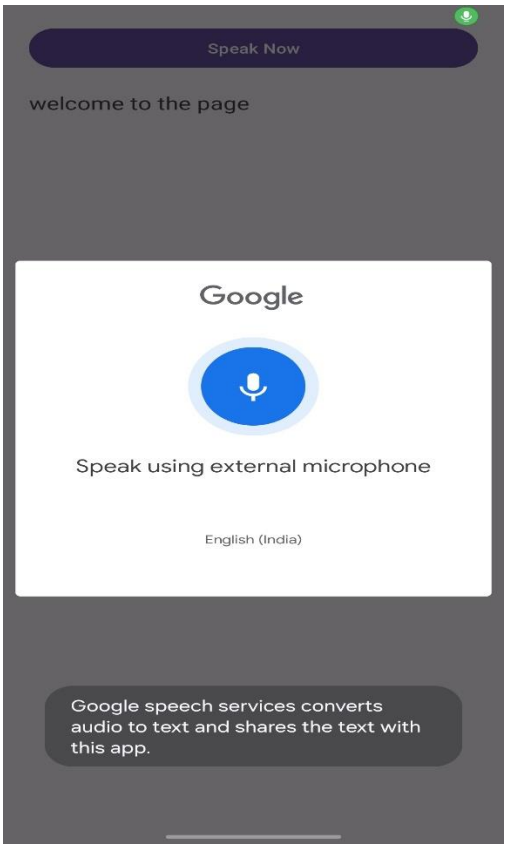
```
    tv.text = result?.get(0)
    }
    }
    }
```

2. AndroidManifest.xml

```
    <manifest package="com.example.exp6"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <uses-permission android:name="android.permission.RECORD_AUDIO"/>
    <application>
    <activity android:name=".MainActivity"/>
</application>
</manifest>
```

OUTPUT:

Speak Now
Converted text will appear here


Speak Now
welcome to the page
Google
Speak using external microphone
English (India)
Google speech services converts audio to text and shares the text with this app.

Speak Now
welcome to the page

1. MainActivity.kt

```
package com.example.smsalertapp
import android.Manifest
import android.content.pm.PackageManager
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
// Request SMS permission
if (ActivityCompat.checkSelfPermission(this, Manifest.permission.RECEIVE_SMS)
!= PackageManager.PERMISSION_GRANTED) {
ActivityCompat.requestPermissions(this,arrayOf(Manifest.permission.RECEIVE_SMS),1)
}
}
}
```

2. SmsReceiver.java

```
package com.example.smsalertapp;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.telephony.SmsMessage;
import android.app.AlertDialog;
public class SmsReceiver extends BroadcastReceiver {
@Override
public void onReceive(Context context, Intent intent) {
```

```
Bundle bundle = intent.getExtras();
if (bundle != null) {
Object[] pdus = (Object[]) bundle.get("pdus");
for (Object pdu : pdus) {
SmsMessage sms = SmsMessage.createFromPdu((byte[]) pdu);
String message = sms.getMessageBody();
AlertDialog.Builder builder = new AlertDialog.Builder(context);
builder.setTitle("New SMS");
builder.setMessage(message);
builder.setPositiveButton("OK", null);
builder.show();
}
}
}
}
package com.example.smsalertapp;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.telephony.SmsMessage;
import android.app.AlertDialog;
public class SmsReceiver extends BroadcastReceiver {
@Override
public void onReceive(Context context, Intent intent) {
Bundle bundle = intent.getExtras();
if (bundle != null) {
Object[] pdus = (Object[]) bundle.get("pdus");
for (Object pdu : pdus) {
SmsMessage sms = SmsMessage.createFromPdu((byte[]) pdu);
String message = sms.getMessageBody();
```

```
AlertDialog.Builder builder = new AlertDialog.Builder(context);
builder.setTitle("New SMS");
builder.setMessage(message);
builder.setPositiveButton("OK", null);
builder.show();    }
}
}
}
```

3. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent">
<TextView
android:id="@+id/textView"
android:layout_centerInParent="true"
android:text="SMS Alert App"
android:textSize="22sp"
android:layout_width="wrap_content"
android:layout_height="wrap_content" />
</RelativeLayout>
```

4. AndroidManifest.xml

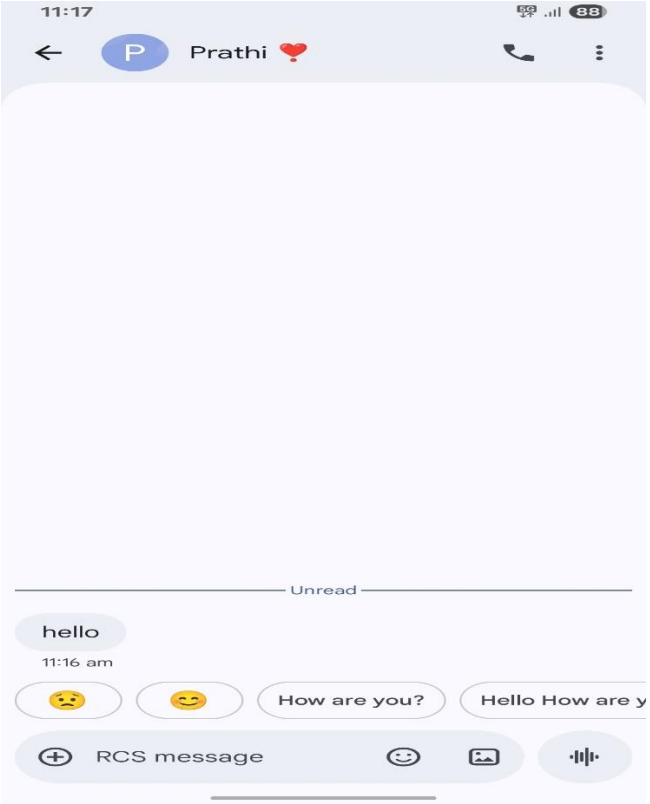
```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.smsalertapp">
<!-- Permissions -->
<uses-permission android:name="android.permission.RECEIVE_SMS"/>
<application
android:allowBackup="true"
android:label="SMSAlertApp"
```

```
android:supportsRtl="true"
android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
 <activity android:name=".MainActivity">
 <intent-filter>
 <action android:name="android.intent.action.MAIN"/>
 <category android:name="android.intent.category.LAUNCHER"/>
 </intent-filter>
 </activity>
 <!-- SMS Receiver -->
 <receiver android:name=".SmsReceiver">
 <intent-filter>
 <action android:name="android.provider.Telephony.SMS_RECEIVED"/>
 </intent-filter>
 </receiver>
 </application>
</manifest>
```

OUTPUT:

SMS Sender App
Enter Phone Number
Enter Message
SEND MESSAGE

SMS Sender App
9842012345
hello
SEND MESSAGE
SMS Sent Successfully


11:17
P
Prathi
Unread
hello
11:16 am
How are you?
Hello How are y
RCS message

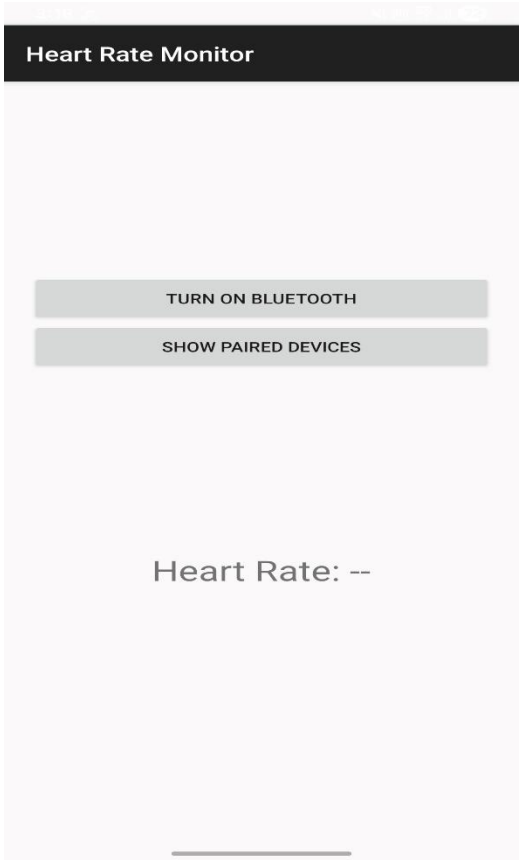
1. MainActivity.kt

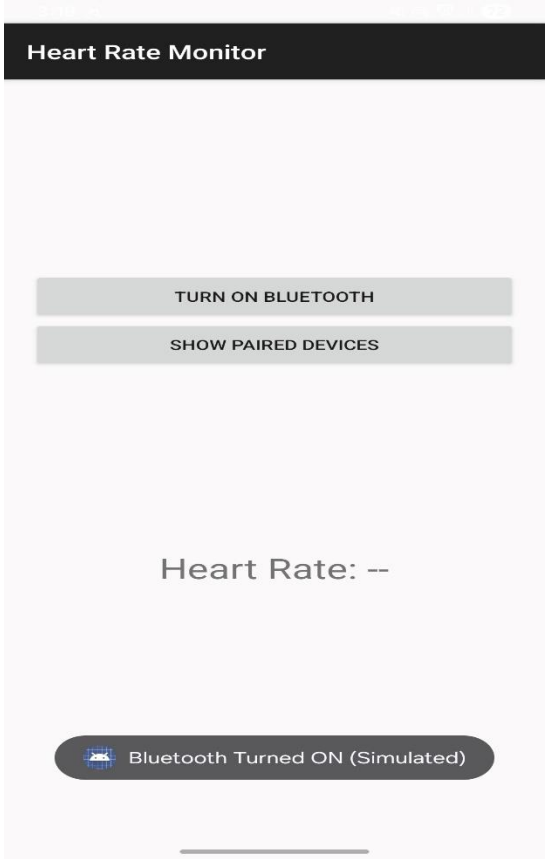
```
package com.example
import android.bluetooth.BluetoothAdapter
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
val tv = TextView(this)
val adapter = BluetoothAdapter.getDefaultAdapter()
tv.text = if (adapter != null && adapter.isEnabled)
"Bluetooth Enabled"
else
"Bluetooth Disabled"
setContentView(tv)
}
}
```

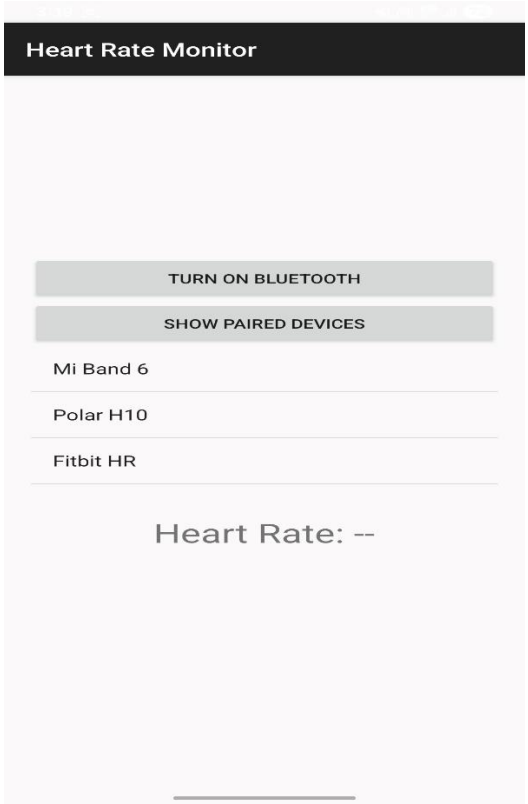
2. AndroidManifest.xml

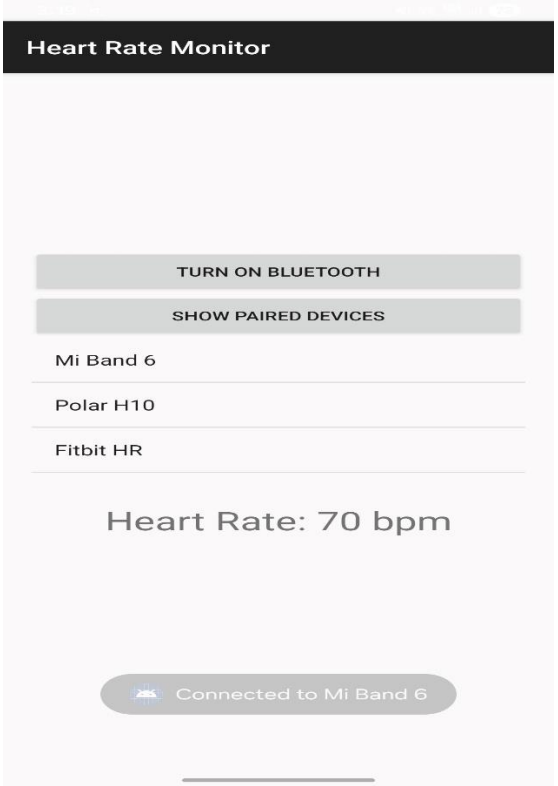
```
<manifest package="com.example.exp8"
xmlns:android="http://schemas.android.com/apk/res/android">
<uses-permission android:name="android.permission.BLUETOOTH"/>
<application>
<activity android:name=".MainActivity"/>
</application>
</manifest>
```

OUTPUT:


Heart Rate Monitor
TURN ON BLUETOOTH
SHOW PAIRED DEVICES
Heart Rate: --


Heart Rate Monitor
TURN ON BLUETOOTH
SHOW PAIRED DEVICES
Heart Rate: --
Bluetooth Turned ON (Simulated)


Heart Rate Monitor
TURN ON BLUETOOTH
SHOW PAIRED DEVICES
Mi Band 6
Polar H10
Fitbit HR
Heart Rate: --


Heart Rate Monitor
TURN ON BLUETOOTH
SHOW PAIRED DEVICES
Mi Band 6
Polar H10
Fitbit HR
Heart Rate: 70 bpm
Connected to Mi Band 6

1. MainActivity.kt

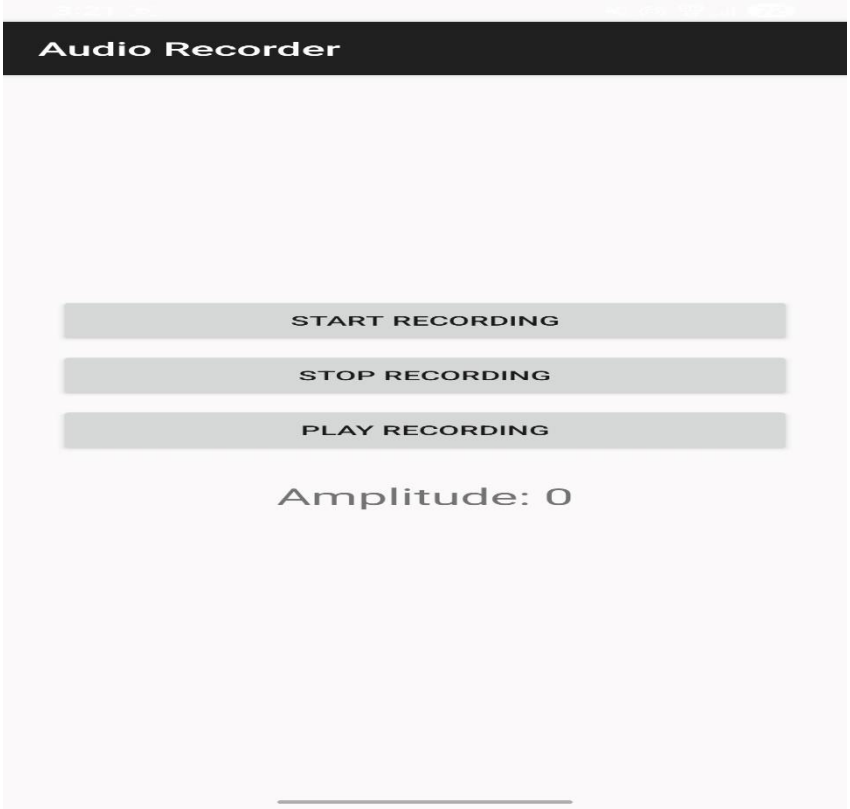
```
package com.example.exp9
import android.media.MediaRecorder
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
var recorder: MediaRecorder? = null
override fun onCreate(savedInstanceState: Bundle?)
super.onCreate(savedInstanceState)
val btn = Button(this)
btn.text = "Record"
setContentView(btn)
btn.setOnClickListener {
recorder = MediaRecorder()
recorder?.setAudioSource(MediaRecorder.AudioSource.MIC)
recorder?.setOutputFormat(MediaRecorder.OutputFormat.THREE_GPP)
recorder?.setAudioEncoder(MediaRecorder.AudioEncoder.AMR_NB)
recorder?.setOutputFile("/sdcard/audio.3gp")
recorder?.prepare()
recorder?.start()
}
}
}
```

2. AndroidManifest.xml

```
<manifest package="com.example.exp9"
xmlns:android="http://schemas.android.com/apk/res/android">
<uses-permission android:name="android.permission.RECORD_AUDIO"/>
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
```

```
<application>
<activity android:name=".MainActivity"/>
</application>
</manifest>
```

OUTPUT:


Audio Recorder
START RECORDING
STOP RECORDING
PLAY RECORDING
Amplitude: 0

Audio Recorder
START RECORDING
STOP RECORDING
PLAY RECORDING
Amplitude: 2665
Recording Started

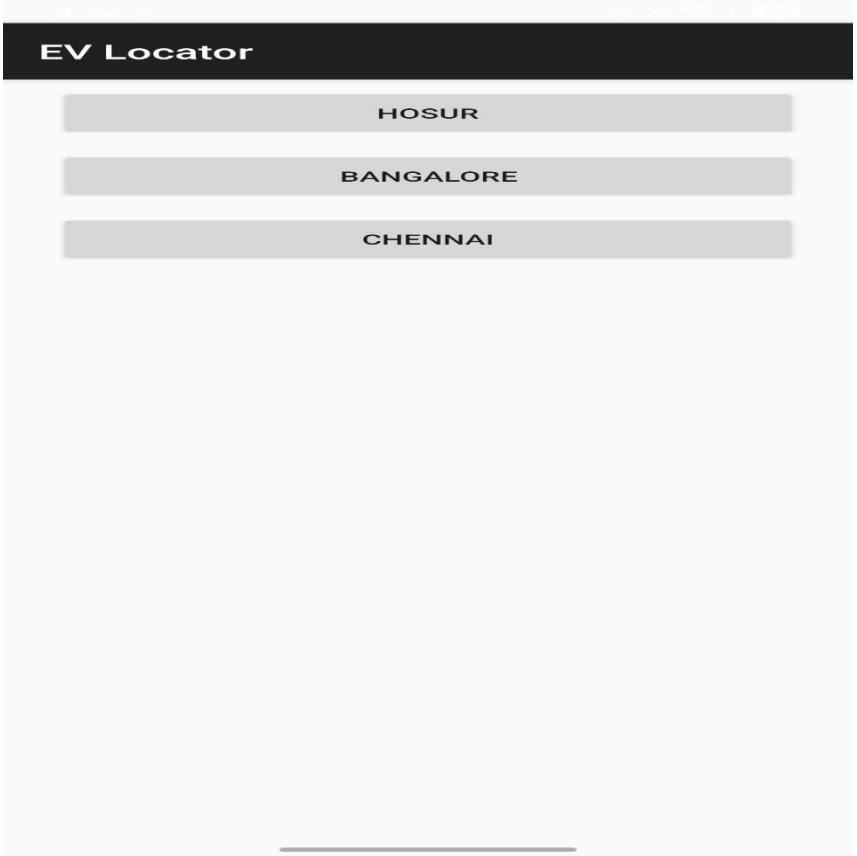
1. MainActivity.kt

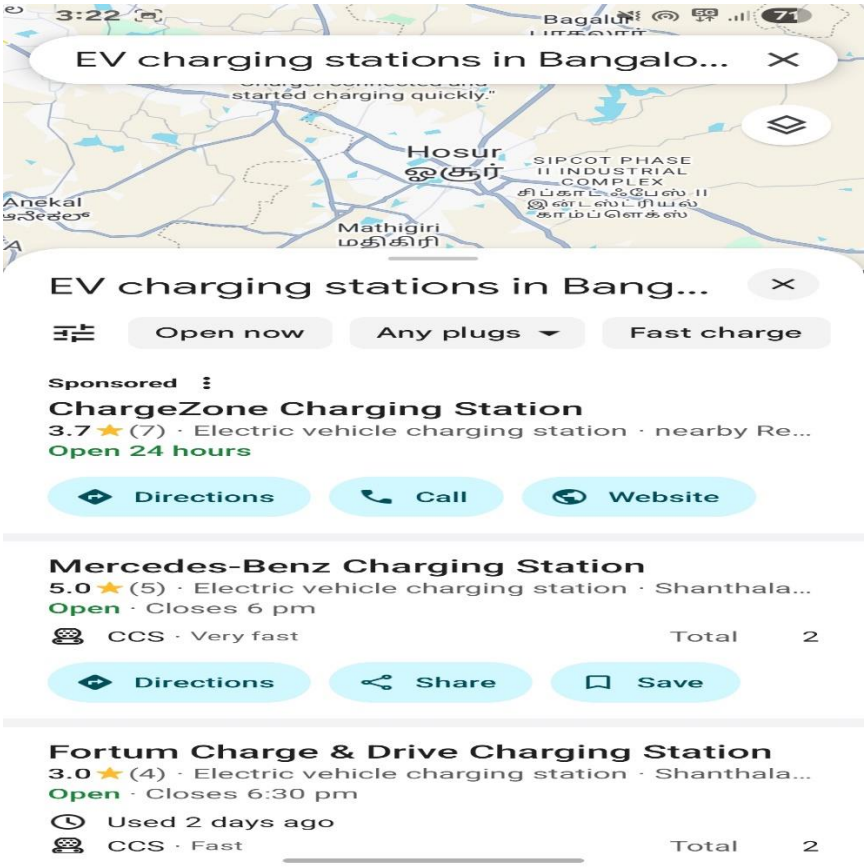
```
package com.example.exp10
import android.content.Intent
import android.net.Uri
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
val btn = Button(this)
btn.text = "Find EV Stations"
setContentView(btn)
btn.setOnClickListener {
val uri = Uri.parse("geo:0,0?q=EV Charging Station")
startActivity(Intent(Intent.ACTION_VIEW, uri))
}
}
}
```

2. AndroidManifest.xml

```
<manifest package="com.example.exp10"
xmlns:android="http://schemas.android.com/apk/res/android">
<application>
<activity android:name=".MainActivity"/>
</application>
</manifest>
```

OUTPUT:


EV Locator
HOSUR
BANGALORE
CHENNAI


3:22
EV charging stations in Bangalo...
Hosur
SIPCOT PHASE II INDUSTRIAL COMPLEX
Anekal
Mathigiri
EV charging stations in Bang...
Open now
Any plugs
Fast charge
Sponsored
ChargeZone Charging Station
3.7 (7) · Electric vehicle charging station · nearby Re...
Open 24 hours
Directions
Call
Website
Mercedes-Benz Charging Station
5.0 (5) · Electric vehicle charging station · Shanthala...
Open · Closes 6 pm
CCS · Very fast
Total 2
Directions
Share
Save
Fortum Charge & Drive Charging Station
3.0 (4) · Electric vehicle charging station · Shanthala...
Open · Closes 6:30 pm
Used 2 days ago
CCS · Fast
Total 2